

Python Basics & Strings

Post-Assessment 1 · 13 questions · 48 points

Name: _____ Date: _____ Score: / 48

This is a **post-assessment**. It shows what you learned in this unit. Try every question, and check your answers if you have time.

Work on your own, without a computer. You have about **45 minutes**. When a question asks what a program prints, write exactly what Python would print, one line of output per answer line. If the code would crash, write the name of the error (for example, `TypeError`). Show your thinking; partial credit is given.

1. Predict the output

3 points

Write exactly what this program prints. Use one answer line for each line of output.

```
print("Ready\nSet")
print('Let\'s go')
print("Age:", 12)
```

2. What type is it?

4 points

Write the type Python would report with `type()`: **int**, **float**, or **str**.

Value	Type
7	
2.0	
"hello"	
8 / 4	

3. Good names, bad names

2 points

Circle every name that is a **legal** Python variable name. Cross out the ones that are not.

top-score

dog_years

1st_try

my name

score2

4. Number crunching

5 points

Write the value of each expression.

Expression	Value	Expression	Value
23 // 4		10 - 2 * 3	
23 % 4		(10 - 2) * 3	
3 ** 2			

5. f-string output

2 points

```
pet = "Biscuit"
kind = "dog"
age = 4
print(f"{pet} the {kind} is {age * 7} in dog years!")
```

What does the last line print?

6. The tip bug

4 points

```
price = input("How much was the pizza? ")
tip = price * 0.2
print(f"Leave a tip of ${tip}")
```

(a) The user types 15. What happens when line 2 runs? Circle one.

(1 pt)

☐ (A) It prints Leave a tip of \$3.0

☐ (B) It crashes with a `TypeError`

☐ (C) It prints Leave a tip of \$15

☐ (D) It prints Leave a tip of \$0.2

(b) Explain **why** in one sentence.

(1 pt)

(c) Rewrite line 1 so the program works (the price might have cents, like 12.50).

(2 pts)

7. Picking characters

4 points

`word = "CASTLE"`. Write what each expression gives. If it causes an error, write the error name.

Expression	Result
<code>len(word)</code>	
<code>word[1]</code>	
<code>word[-2]</code>	
<code>word[10]</code>	

8. Slice it

4 points

`text = "rainbow"`. Write the string each slice gives.

Slice	Result
<code>text[0:4]</code>	
<code>text[4:7]</code>	
<code>text[2:5]</code>	
<code>text[::-1]</code>	

9. Can you change a string?

2 points

```
name = "jake"
name[0] = "J"
```

(a) What happens when line 2 runs? Circle one.

(1 pt)

☐ (A) name becomes "Jake"

☐ (B) name becomes "Jjake"

☐ (C) Python shows an error because strings can't be changed

☐ (D) Nothing happens

(b) Write one line of code that makes `name` equal `"Jake"` by building a new string.

(1 pt)

10. Dot magic

5 points

`s = "happy birthday"`. Write the result of each expression.

Expression	Result
<code>s.upper()</code>	
<code>s.title()</code>	
<code>s.count("p")</code>	
<code>s.replace("p", "b")</code>	
<code>" go ".strip()</code>	

11. Text toolbox

5 points

`sentence = "birds eat seeds"`. Write the result of each expression.

Expression	Result
<code>sentence.split()</code>	
<code>"+".join(["x", "y", "z"])</code>	
<code>sentence.find("eat")</code>	
<code>sentence.startswith("bird")</code>	
<code>"fish" in sentence</code>	

12. Nickname Maker

3 points

Write a program that asks the user for their name, then prints the name in Title Case on one line and a nickname made of the **first three letters** of the Title Case name on the next line.

EXAMPLE RUN

```
Name: sAMANTHA
Samantha
Sam
```

13. Age in Dog Years

5 points

Write a program that:

- asks for the pet's name
- asks for the pet's age in years (a whole number)
- works out its age in dog years ($\text{age} \times 7$)
- prints a message like the one below using an **f-string**

EXAMPLE RUN

```
Pet name: Biscuit
Age: 4
Biscuit is 28 in dog years!
```

END OF ASSESSMENT